

FRONT-END DEVELOPMENT HACKATHON Level-2

Activity Tracker with Real-Time Progress

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TITLE : Activity Tracker with Real-Time Progress
GITHUB LINK :

1. Objective

The objective of this project is to develop an interactive activity tracker that allows users to mark tasks as completed and view their progress in real time. The application uses JavaScript to update the user interface dynamically without refreshing the page.

2. Problem Focus

In the previous level, the application had a static user interface where users could only view tasks but could not interact with them. This project solves that problem by converting the static UI into a dynamic one, allowing user interaction and real-time updates.

3. Application Overview

This application is an activity tracker that helps users manage their daily tasks. Users can view activities, mark them as completed, and track their progress continuously.

4. Data Handling Approach

The application stores task data using a JavaScript array, where each task contains a name and its completion status. This makes it easy to manage and update tasks. Local storage can also be used to save user progress.

5. UI Components Design

The user interface includes an activity list, buttons to mark tasks as completed, and a progress display section. These components provide a simple and user-friendly experience.

6. Interactivity Implementation

Interactivity is implemented using JavaScript event handling. When a user clicks a button, the task status is updated immediately, and the interface reflects the changes without reloading the page.

7. Application Logic

The application logic counts the total number of tasks and the number of completed tasks. It then calculates the progress and updates the display dynamically.

8. Technology Used

This project is developed using HTML for structuring the webpage, CSS for styling, and JavaScript for adding functionality and interactivity.

9. Best Practices Followed

The project follows good coding practices such as using meaningful variable names, avoiding hardcoding, adding comments, and handling possible errors effectively.

10. Key Features Implemented

The main features include task completion, real-time progress tracking, and dynamic updates of the user interface.

11. Expected Output / Deliverables

The final output is a fully functional activity tracker with an interactive interface, real-time progress display, and a working implementation shared through a GitHub repository.

12. Learning Outcome

Through this project, important concepts like event handling, DOM manipulation, and building interactive web applications have been learned.

13. Conclusion

This project successfully converts a static user interface into an interactive Activity Tracker. Users can mark tasks as completed, and the application updates progress in real-time without page reload. By using JavaScript for event handling and DOM manipulation, the app

becomes dynamic and user-friendly. Overall, this project enhances user experience and demonstrates practical front-end development skills.

14. Coding

HTML

```
<!DOCTYPE html>

<html lang="en">

<head>

  <meta charset="UTF-8">

  <title>Activity Tracker</title>

  <link rel="stylesheet" href="style.css">

</head>

<body>

<div class="container">

  <h1>Activity Tracker</h1>

  <p id="progress">0 out of 0 completed</p>

  <ul id="activityList"></ul>

</div>

<script src="script.js"></script>

</body>

</html>
```

CSS

```
body {

  font-family: Arial, sans-serif;

  background-color: yellow;
```

```
    text-align: center;
}

.container {
    width: 400px;
    margin: 50px auto;
    background: white;
    padding: 20px;
    border-radius: 10px;
    box-shadow: 0 0 10px gray;
}

h1 {
    color: #333;
}

ul {
    list-style: none;
    padding: 0;
}

li {
    background: #eaeaea;
    margin: 10px 0;
    padding: 10px;
    border-radius: 5px;
    display: flex;
    justify-content: space-between;
```

```
    align-items: center;
}
button {
    padding: 5px 10px;
    border: none;
    background-color: green;
    color: white;
    border-radius: 5px;
    cursor: pointer;
}
button:hover {
    background-color: darkgreen;
}
.completed {
    text-decoration: line-through;
    color: gray;
}
```

JAVA SCRIPT

```
// Activity Data
let activities = [
    { name: "HTML Practice", completed: false },
    { name: "CSS Styling", completed: false },
    { name: "JavaScript Learning", completed: false },
    { name: "Mini Project", completed: false }
```

```
];  
  
// Render Activities  
  
function renderActivities() {  
  
    const list = document.getElementById("activityList");  
  
    list.innerHTML = "";  
  
    let completedCount = 0;  
  
    activities.forEach((activity, index) => {  
  
        const li = document.createElement("li");  
  
        if (activity.completed) {  
  
            li.classList.add("completed");  
  
            completedCount++;  
  
        }  
  
        li.innerHTML = `  
            ${activity.name}  
  
            <button onclick="markComplete(${index})">  
  
                ${activity.completed ? "Done" : "Complete"}  
  
            </button>  
  
        `;  
  
        list.appendChild(li);  
  
    });  
  
    // Update Progress  
  
    document.getElementById("progress").innerText =  
  
    `${completedCount} out of ${activities.length} completed`;  
  
    // All Completed Message
```

```
if (completedCount === activities.length) {  
    document.getElementById("progress").innerText += " 🎉 All Activities Completed!";  
}  
}  
  
// Mark Complete Function  
function markComplete(index) {  
    activities[index].completed = true;  
    renderActivities();  
}  
  
// Initial Load  
renderActivities();
```

15. Output

